

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EC)

April/May 2012

EC 303 Audio Video Engineering

Date: 02.05.2012, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Write the Requisites of good microphone. [04]
(b) Why cone type speaker is known as direct radiating type? [03]
- Q - 2 (a) Explain the directivity of Ribbon Microphone. [03]
(b) Calculate the efficiency of loudspeaker if power intensity of sound is 0.08 W/m^2 at 1 meter distance from the loudspeaker, when the amplifier delivers 25 watt. [04]
(c) Explain growth and decay of sound in an Enclosure. [07]

OR

- Q - 2 (a) Explain Electrodynamic Loudspeaker. [03]
(b) Explain Moving coil Microphone. [04]
(c) There is a hall of 20m x 15m x 5m having following surfaces. [07]

Absorbent material	Dimension		Absorption coefficient
Open windows	2m x 1m	8 Nos	1
Thin carpet	15m x 10m	1 Nos	0.25
Woolen Galeecha	10m x 5m	1 Nos	0.58
Curtains	2.5m x 1m	8 Nos	0.15
Audience		410	0.45

Calculate reverberation time.

- Q - 3 (a) Draw and Explain basic PA System. [05]
(b) Define Hypermedia and multimedia with suitable example. [07]
(c) Define (i) Hue (ii) Brightness [02]

OR

- Q - 3 (a) Two loudspeakers of 8 ohm each and 4 loudspeakers of 4 ohm each are to be connected to an amplifier of 8 ohm output. How Will you Connect? [05]
- (b) List the applications of Multimedia and explain any one. [07]
- (c) Why synchronization pulses are required in TV system? [02]

SECTION – II

- Q - 4 (a) What do you understand by interlaced scanning? Explain interlaced scanning in brief [05]
- (b) Justify the choice of 625 lines for TV transmission [02]
- Q - 5 (a) Sketch composite video signal waveform for at least three successive lines and indicate: extreme white level, blanking level, pedestal height and sync pulse level. [07]
- (b) Show that total channel Bandwidth of 11.25MHz would be necessary if both sidebands of modulated picture signal are fully transmitted. Explain how the channel width is reduced to 7MHz by allowing vestigial sideband transmission. [07]

OR

- Q - 5 (a) What do you understand by positive and negative modulation? List advantages and disadvantages of each with respect to TV transmission. [07]
- (b) Why TV Transmission is done at frequencies above 40MHz? Justify that space wave propagation is the only effective mode for TV signal radiation above about 40MHz [07]
- Q - 6 (a) Explain VHF tuner with block diagram [05]
- (b) Draw and explain elementary block diagram of a monochrome TV transmitter [05]
- (c) Explain SAW filter in brief [04]

OR

- Q - 6 (a) Write a short note on Interferences suffered by TV signal [05]
- (b) Explain high definition TV(HDTV) [05]
- (c) Explain the three dimensional (3D) Television. [04]
